COS326 November 2020 Examination

Test Information

Description	**University of Pretoria, School for Information Technology, EBIT Faculty** **COS 326 Database Systems, November 2020 Examination** **Examiner:** Dr. Patricia E.N. Lutu **External Examiner:** Prof. Roelien Goede **Date:** 17 November 2020 **TIME:** 2 hours 30 minutes **Marks:** 90
Instructions	**ATTEMPT ALL QUESTIONS**
Timed Test	This test has a time limit of 2 hours and 30 minutes.This test will save and be submitted automatically when the time expires. Warnings appear when **half the time**, **5 minutes**, **1 minute**, and **30 seconds** remain.*[The timer does not appear when previewing this test]*
Multiple Attempts	Not allowed. This Test can only be taken once.
Force Completion	This Test can be saved and resumed at any point until time has expired. The timer will continue to run if you leave the test.
	Your answers are saved automatically.

Question Completion Status:

QUESTION 1

1. State whether the following statement about the MapReduce computational framework is TRUE or FALSE:

   The MapReduce computational framework enables the implementation of highly distributed programs that run on failure-tolerant and scalable clusters of commodity machines.

   True

   False

1 points

QUESTION 2

1. Why are NoSQL databases suitable for the storage of Big Data?

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Paragraph
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10pt
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0 WORDSPOWERED BY TINY

3 points

QUESTION 3

1. One option for storing XML document data in a database is to store the XML document as a value of one row and one column of a table/relation.

   a. Give a concise description of this approach.
   b. Discuss the strengths and weaknesses of this approach.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).

Paragraph
Open Sans,arial,helvetica,sans-serif
10pt
P
0 WORDSPOWERED BY TINY

3 points

QUESTION 4

1. State whether the following statement is TRUE or FALSE:

   Three characteristics of Big Data are big volume, big velocity and big verification.

   True

   False

1 points

QUESTION 5

1. State whether the following statement is TRUE or FALSE:

   A NoSQL database is a distributed non-relational database system designed for large-scale data storage and for massively parallel data processing using powerful server machines.

   True

   False

1 points

QUESTION 6

1. State whether the following statement about the MapReduce computational framework is TRUE or FALSE:

   There are two phases in a map-reduce computation: map and reduce.

   True

   False

1 points

QUESTION 7

1. State whether the following statement about Object-relational databases (ORDB) is TRUE or FALSE:

   ORDBs provide the object-oriented extensions to SQL, to database users. These extensions enable database users to implement user-defined types and inheritance.

   True

   False

1 points

QUESTION 8

1. State whether each of the following statement about OODBMSes is TRUE or FALSE:

   Objects may be organised into groups and one special object is responsible for storing the class methods on behalf of all the other objects in the group.

True

False

1 points

QUESTION 9

1. Suppose the following SQL statements have been executed on a PostgreSQL database:

```
CREATE SEQUENCE venueSeq START 101;
CREATE TYPE BuildingCodeType AS ENUM ('IT', 'LAW', 'CHM', 'PHY');
CREATE TYPE RoomCodeType AS (building BuildingCodeType, floor int, room int);
CREATE TYPE EquipmentType AS ENUM ('projector', 'PAsystem', 'safe', 'PC',
'phone');

CREATE TABLE Venue  (
        venueKey     integer DEFAULT nextval('venSeq') PRIMARY KEY,
          venueCode    RoomCodeType,
          seats  int  );

CREATE TABLE LectureRoom  (
          equipmentList  text [ ] )
          INHERITS (VENUE);
```

It is now required to insert the following data into the LectureRoom table.

venue key: auto-generated
venue code: IT 4-2
seats: 120
equipment: projector, PAsystem, safe

Which one of the following SQL statements will correctly insert the data into the LectureRoom table?

a. INSERT INTO LectureRoom (venueCode, seats, equipmentList)
VALUES (('IT', 4,2), 120, ARRAY['projector', 'PAsystem', 'safe']);
b. INSERT INTO LectureRoom (venueCode, seats, equipmentList)
VALUES (('IT', 4,2), 120, ['projector', 'PAsystem', 'safe']);
c. INSERT INTO LectureRoom (venueCode, seats, equipmentList)
VALUES (('IT', 4,2), 120, ('projector', 'PAsystem', 'safe'));

3 points

QUESTION 10

1. State whether the following statement about the MongoDB document database is TRUE or FALSE:

Two approaches to database design are: *embedded data model* and *normalized data model.*

True

False

1 points

QUESTION 11

1. State whether the following statement about the MongoDB document database is TRUE or FALSE:

The primary construct used in the specification of queries and returning query results is the JSON object.

True

False

1 points

QUESTION 12

1. State whether the following statement about New SQL database management systems is TRUE or FALSE:

   New SQL DBMSes support the relational data model and SQL as the primary mechanism for application interaction.

   True

   False

1 points

QUESTION 13

1. State whether the following statement about Object-oriented database management systems (OODBMS) is TRUE or FALSE:

   OODBMSs use object identifiers (OIDs) to uniquely identify database objects and to represent one-to-one, one-to-many, many-to-many relationships between database objects.

   True

   False

1 points

QUESTION 14

1. State whether the following statement about New SQL database management systems is TRUE or FALSE:

   New SQL DBMSes aim to have the same level of scalability and availability as NoSQL DBMSes and maintain the ACID properties.

   True

   False

1 points

QUESTION 15

1. Given that the following Cypher query has been executed on a Neo4j graph database:

```
CREATE
      ( psT:Person { name: "Thandi", from: "Durban",  hobby: "singing" }  ),
       ( psJ:Person { name: "Johan", from: "Pretoria",  hobby: "surfing" }   ),
       (psN:Person { name: "Neo", from: "Tshwane", hobby: "soccer"   }  ),

       (psT)-[:FOLLOWS { since: 2017 } ]->(psJ ),
       (psT)-[:FOLLOWS { since: 2016 } ]->(psN ),
       (psJ)-[:FOLLOWS  { since: 2018 } ]->(psT )
      (psJ)-[:FOLLOWS  { since: 2018 } ]->(psN )
```

   Which one of the following queries will display the details of the persons that Johan follows?

a. MATCH (psT {name: "Johan"})-[:FOLLOWS]-> (followed)
RETURN name, from, hobby;

b. MATCH (psT {name: "Johan"})-[:FOLLOWS]-> (followed)
RETURN followed.name, followed.from, followed.hobby;

c. MATCH (psT {name = "Johan"})-[:FOLLOWS]-> (followed)
RETURN psT.name, psT.from, psT.hobby;

3 points

QUESTION 16

1. State whether the following statement about graph databases is TRUE or FALSE:

Three examples of application domains for graph databases are: representing social networks, generating recommendations, and conducting forensic investigations.

True

False

1 points

QUESTION 17

1. State whether the following statement about the data mining is TRUE or FALSE:

Three modeling techniques that are commonly used data mining are: descriptive modeling, predictive modeling, and prescriptive modeling.

True

False

1 points

QUESTION 18

1. State whether the following statement about graph databases is TRUE or FALSE:

The data in a graph database is stored using an adjacency matrix where each cell in the matrix stores a *key-value* pair for the relationship between two graph nodes.

True

False

1 points

QUESTION 19

1. State whether the following statement about OODBMSes is TRUE or FALSE:

The class methods may be stored in external files, and then the libraries are linked into the application at compile / link time.

True

False

1 points

QUESTION 20

1. Given the following Java program declarations for a db4o object oriented database:

```
final static String DB4OFILENAME = System.getProperty("user.home")  +
"/Driver.db4o";
ObjectContainer db =
                Db4oEmbedded.openFile(Db4oEmbedded.newConfiguration(), DB4OFILEN
AME);
```

Which one of the following pieces of Java code will perform an update operation?

a.
```
ObjectSet result =  db.queryByExample(new Driver ("Lewis Hamilton", 0));
Driver found = (Driver) result.next();
found.addPoints(10);
db.store(Driver);
```

b.
```
ObjectSet result =  db.queryByExample(new Driver ("Lewis Hamilton", 0));
Driver found = (Driver) result.next();
found.addPoints(10);
db.store(found);
```

c.
```
ObjectSet result =  db.query(new Driver ("Lewis Hamilton", 0));
Driver found = (Driver) result.next();
found.addPoints(10);
db.store(found);
```

3 points

QUESTION 21

1. Explain the similarities and differences between the data models for NoSQL databases which are classified as **key-value stores** and NoSQL databases which are classified as **wide-column stores**.

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Paragraph
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10pt
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0 WORDSPOWERED BY TINY

5 points

QUESTION 22

1. State whether the following statement about Object-oriented database management systems (OODBMS) is TRUE or FALSE:

   One strategy for developing an OODBMS is to integrate the SQL execution engine into the compiler of an OOP language.

   True

   False

1 points

QUESTION 23

1. State whether the following statement is TRUE or FALSE:

   Four categories of NoSQL databases are: key-value stores, document databases, wide-column stores, and graph databases.

   True

   False

1 points

QUESTION 24

1. State whether the following statement about NoSQL database systems is TRUE or FALSE:

   All NoSQL databases use the Big Data Model (BDM).

   True

   False

1 points

QUESTION 25

1. State whether the following statement about NoSQL database systems is TRUE or FALSE:

   All NoSQL databases provide Atomicity, Consistency, Isolation, Durability (ACID) guarantees for database transactions.

   True

   False

1 points

QUESTION 26

1. State whether the following statement about New SQL database management systems is TRUE or FALSE:

   New SQL DBMSes provide a brand new query language called New SQL.

   True

   False

1 points

QUESTION 27

1. State whether the following statement about the MapReduce computational framework is TRUE or FALSE:

   The MapReduce computational framework is one of the solutions for conducting fast computations in Big Data environments.

   True

   False

1 points

QUESTION 28

1. State whether the following statement about the MongoDB document database is TRUE or FALSE:

   Three types of queries for aggregation are: pipeline, map-reduce, and single purpose aggregation queries.

   True

   False

1 points

QUESTION 29

1. State whether the following statement about the data mining is TRUE or FALSE:

Data mining is the process of extracting data from transactional databases, cleaning the data, and storing the clean data back into the database.

True

False

1 points

QUESTION 30

1. State whether the following statement about NoSQL database systems is TRUE or FALSE:

   Some NoSQL databases provide Atomicity, Consistency, Isolation, Durability (ACID) guarantees for database transactions while others only provide guarantees of eventual consistency.

True

False

1 points

QUESTION 31

1. Another option for storing XML document data in a database is to store the XML document in a *shredded* form.

   a. Give a concise description of this approach.
   b. Discuss the strengths and weaknesses of this approach.

For the toolbar, press ALT+F10 (PC) or ALT+FN+F10 (Mac).
Paragraph
Open Sans,arial,helvetica,sans-serif
10pt
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0 WORDSPOWERED BY TINY

3 points

QUESTION 32

1. State whether the following statement about graph databases is TRUE or FALSE:

   A graph is represented as an object-oriented network of nodes (objects), edges (objects for node relationships) and properties which represent object attributes expressed as *key-value* pairings.

True

False

1 points

QUESTION 33

1. Given the following contents of the collection *blogsCollection* in the MongoDB database *mydb*:

```
{
        _id:            ObjectId(7df78ad8902c)
        title:           'Cassandra Overview',
        description:     'Cassandra is a NoSql database',
        tags:           ['Cassandra', 'database', 'NoSQL'],
        likes:          100
},
```

```
{
        _id:            ObjectId(7df78ad8902d)
        title:          'NoSQL Overview',
        description:    'NoSql database is very fast',
        tags:           ['mongodb', 'neo4j', 'database', 'NoSQL'],
        likes:          200
},
{
        _id:            ObjectId(7df78ad8902e)
        title:          'VoltDB Overview',
        description:    'VoltDB is a New sql database',
        tags:           ['VoltDB', 'database', 'New SQL'],
        likes:          750
}
```

It is now required to write the JavaScript statement for the *mongo* shell to compute and display the total number of *likes* for all documents in the collection *blogsCollection* with the title 'NoSQL Overview'.

Which of the statements below will perform the required task?

a.
```
use mydb
db. blogsCollection.aggregate(
    [
        {  $match: {title:"NoSQL Overview"}  },
        {  $group:
            {_id: "$title", num_likes: { $sum : "$likes"} }
        }
    ] )
```

b.
```
use mydb
db. blogsCollection.aggregate(
    [
        {  $match: {title:"NoSQL Overview"}  },
        {  $group by:
            {_id: "$title", num_likes: { $sum : "$likes"} }
        }
    ] )
```

c.
```
use mydb
db. blogsCollection.aggregate(
    [
        {  $match: {title:"NoSQL Overview"}  },
        {  $group:
            {_id: "$title", num_likes: { $add : "$likes"} }
        }
    ] )
```

3 points

QUESTION 34

1. The list below gives two of the limitations associated with relational database systems

   (i) Due to normalisation, information is obtained from the database through performing expensive joins of 2 or more tables.

   (ii) Schema changes are difficult. Database schema changes require program changes.

   Explain how the MongoDB NoSQL database avoids the above weaknesses.

Paragraph
Open Sans,arial,helvetica,sans-serif
10pt
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0 WORDSPOWERED BY TINY

4 points

QUESTION 35

1. The manager of a business division wants an answer to the following question: "How much revenue did the gysmo product generate by province and by month this year?"
   Which one of the following SQL queries will produce the required information?

a.
```
SELECT B.province, T.month, SUM(S.revenue) AS gysmoRevenue
FROM BRANCH B, TIME T, PRODUCT P, SALES S
WHERE P.ProductName = 'gysmo'        AND
      T.year = 2020                  AND
      B.BranchKey = S.BranchKey      AND
      T.TimeKey    = S.TimeKey       AND
      P.ProductKey = S.ProductKey
GROUP ON CUBE (B.province, T.month);
```

b.
```
SELECT B.province, T.month, SUM(S.revenue) AS gysmoRevenue
FROM BRANCH B, TIME T, PRODUCT P, SALES S
WHERE P.ProductName = 'gysmo' AND
      T.year = 2020                  AND
      B.BranchKey = S.BranchKey      AND
      T.TimeKey    = S.TimeKey       AND
      P.ProductKey = S.ProductKey
GROUP BY ROLLUP (B.province, T.month);
```

c.
```
SELECT B.province, T.month, SUM(S.revenue) AS gysmoRevenue
FROM BRANCH B, TIME T, PRODUCT P, SALES S
WHERE P.ProductName = 'gysmo' AND
      T.year = 2020                  AND
      B.BranchKey = S.BranchKey      AND
      T.TimeKey    = S.TimeKey       AND
      P.ProductKey = S.ProductKey
GROUP BY FIELDS (B.province, T.month);
```

3 points

QUESTION 36

1. State whether the following statement about Object-oriented databases (OODB) is TRUE or FALSE:
   An OODB is a persistent sharable collection of objects defined by an object-oriented data model.

True

False

1 points

QUESTION 37

1. Three of the supporting technologies for the New SQL database management approach are:
   1. column-oriented data storage
   2. in-memory processing
   3. massively parallel processing

Provide a brief explanation of how each of the above technologies supports the New SQL database management approach.

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Paragraph
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10pt
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0 WORDSPOWERED BY TINY

6 points

QUESTION 38

1. a. Give a definition of data mining.

   b. Briefly describe the phases of the Cross- Industry Standard Process for Data Mining (CRISP-DM) for data mining projects. Explain why each phase is important.

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Paragraph
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10pt
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0 WORDSPOWERED BY TINY

6 points

QUESTION 39

1. The PostgreSQL database table below has been created using the following statement:
   CREATE TABLE XMLStaff (docNo CHAR(4), staffData XML);

The data shown below has then imported into the table.

docNo	staffData
1	<STAFF branchNo = "B001"> <STAFFNO>S010</STAFFNO> <NAME> <FNAME>Anna</FNAME> <LNAME>Brown</LNAME> </NAME> </STAFF>
2	<STAFF branchNo = "B002"> <STAFFNO>S011</STAFFNO> <NAME> <FNAME>Thabiso</FNAME> <LNAME>Mashaba</LNAME> </NAME> </STAFF>
3	<STAFF branchNo = "B002"> <STAFFNO>S012</STAFFNO> <NAME> <FNAME>Jaco</FNAME> <LNAME>van Wyk</LNAME> </NAME> </STAFF>

Which one of the following PostgreSQL statements will display the staff number and name for all staff, in the following format?

StaffNumber	FirstName	Surname
S010	Anna	Brown
S011	Thabiso	Mashaba
S012	Jaco	van Wyk

a.
```
SELECT xpath('/STAFF/STAFFNO/text()', staffData)  AS StaffNumber,
       xpath ('/STAFF/NAME/FNAME/text()', staffData)  AS FirstName,
       xpath('/STAFF/NAME/LNAME/text()', staffData)  AS Surname
FROM   XMLStaff;
```

b.
```
SELECT xpath('/STAFF/STAFFNO', staffData)  AS StaffNumber,
       xpath('/STAFF/NAME/FNAME', staffData)  AS FirstName,
       xpath('/STAFF/NAME/LNAME', staffData)  AS Surname,
 FROM   XMLStaff;
```

c.
```
SELECT unnest (xpath('/STAFF/STAFFNO/text()', staffData) ) AS
       StaffNumber,
       unnest (xpath('/STAFF/NAME/FNAME/text()', staffData) ) AS
       FirstName,
       unnest (xpath('/STAFF/NAME/LNAME/text()', staffData) ) AS
       Surname,
 FROM   XMLStaff;
```

3 points

QUESTION 40

1. State whether the following statement about the data mining is TRUE or FALSE:

   Sentiment analysis of microblogs (e.g. tweets) can be conducted by storing the data in a NoSQL database and using text mining methods to determine the sentiment for each microblog.

   True

   False

1 points

QUESTION 41

1. **Note: Due to a technical problem, the dollar sign is represented as ~~S~~ in this question.**
   Suppose the following SQL statements have been executed on a PostgreSQL database:

```
CREATE TABLE cities  ( name text, population real, area int  );
CREATE FUNCTION check_valid_area() RETURNS TRIGGER AS
~~SS~~
    BEGIN
        IF (NEW.area <= 0) THEN
            RAISE EXCEPTION 'Invalid area value!';
        END IF;
        RETURN NEW;
    END;
~~SS~~ LANGUAGE plpgsql;
```

   Which one of the following statements will create a trigger for the cities table?

a.
```
CREATE TRIGGER valid_area
    BEFORE INSERT  OR UPDATE ON Cities
        FOR EACH ROW
            EXECUTE PROCEDURE check_valid_area();
```

b.
```
CREATE TRIGGER valid_area
    BEFORE INSERT  OR UPDATE ON Cities
            EXECUTE PROCEDURE check_valid_area();
```

c.
```
CREATE TRIGGER valid_area
```

```
FOR EACH ROW
        EXECUTE PROCEDURE check_valid_area();
```

3 points

QUESTION 42

1. State whether the following statement about Object-relational databases (ORDB) is TRUE or FALSE:

   ORDBs reduce the need for database table joins by enabling the storage of objects in tables.

True

False

1 points

QUESTION 43

1. Given the following XML document:

```
<STAFFLIST>
<STAFF branchNo = "B005">
       <STAFFNO>SL21</STAFFNO>
       <NAME>
      <FNAME>John</FNAME><LNAME>Green</LNAME>
       </NAME>
       <SALARY>30000</SALARY>
</STAFF>
<STAFF branchNo = "B003">
       <STAFFNO>SG37</STAFFNO>
       <NAME>
      <FNAME>Ann</FNAME><LNAME>Brown</LNAME>
       </NAME>
       <SALARY>12000</SALARY>
</STAFF>
</STAFFLIST>
```

Which one of the FLWOR queries below will list the staff names and staff numbers of all staff in ascending order of staff number?

a.
```
for $s in //STAFF
let $snum   :=    /STAFFNO
let $fn       :=   /NAME/FNAME
let $ln       :=   /NAME/LNAME
order by     /STAFFNO ascending
return       <STAFFDATA>
                      <STAFFNO>{ $snum/text() }</STAFFNO>
                      <FNAME>{$fn/text() }</FNAME>
                      <LNAME>{$ln/text() }</LNAME>
             </STAFFDATA>
```

b.
```
let $snum   :=   /STAFFNO
let $fn       :=   /NAME/FNAME
let $ln       :=   /NAME/LNAME
order by     /STAFFNO ascending
return       $snum/text(), $fn/text(), $ln/text()
```

c.
```
for $s in //STAFF
let $snum   :=   $s/STAFFNO
let $fn       :=   $s/NAME/FNAME
let $ln       :=   $s/NAME/LNAME
order by $s/STAFFNO ascending
return       <STAFFDATA>
                      <STAFFNO>{$snum/text() }</STAFFNO>
                      <FNAME>{$fn/text() }</FNAME>
```

```
        <LNAME>{$ ln/text() }</LNAME>
    </STAFFDATA>
```

3 points

QUESTION 44

1. **Note: Due to a technical problem, the dollar sign in this question is represented as S**
   Suppose the following SQL statements have been executed on a PostgreSQL database:

```
CREATE SEQUENCE venueSeq START 101;
CREATE TYPE BuildingCodeType AS ENUM ('IT', 'LAW', 'CHM', 'PHY');
CREATE TYPE RoomCodeType AS (building BuildingCodeType, floor int, room int);
CREATE TYPE EquipmentType AS ENUM ('projector', 'PAsystem', 'safe', 'PC',
'phone');

CREATE TABLE Venue  (
        venueKey     integer DEFAULT nextval('venSeq') PRIMARY KEY,
        venueCode    RoomCodeType,
        seats  int  );

CREATE TABLE LectureRoom  (
        equipmentList  text [ ] )
        INHERITS (VENUE);
```

   It is now required to implement the function *hasProjector*, which will return TRUE if a lecture room has a projector or FALSE otherwise. Which one of the following is a correct implementation of the *hasProjector* function?

a.
```
CREATE FUNCTION hasProjector ( text [ ]) RETURNS boolean AS
SS
    DELARE val text; proj boolean;
    BEGIN
            proj = FALSE;
            FOR val IN ARRAY S1
            LOOP
                IF (val = 'projector') THEN  proj = TRUE;
                ENDIF;
            END LOOP
            RETURN  proj;
    END;
SS LANGUAGE plpgsql;
```

b.
```
CREATE FUNCTION hasProjector ( text [ ]) RETURNS boolean AS
SS
    DELARE val text; proj boolean;
    BEGIN
            proj = FALSE;
            FOREACH val IN ARRAY S1
            LOOP
                IF (val = 'projector') THEN  proj = TRUE;
                ENDIF;
            END LOOP
            RETURN  proj;
    END;
SS  LANGUAGE plpgsql;
```

c.
```
CREATE FUNCTION hasProjector ( text ) RETURNS boolean AS
SS
```

```
DELARE val text; proj boolean;
BEGIN
        proj = FALSE;
        FOREACH val IN ARRAY $1
        LOOP
            IF (val = 'projector') THEN  proj = TRUE;   ENDIF;
        END LOOP
        RETURN  proj;
END;
$$  LANGUAGE plpgsql;
```

3 points

QUESTION 45

1. State whether the following statement about Object-relational databases (ORDB) is TRUE or FALSE:

   ORDBs provide a seamless integration between applications and the database.

   True

   False

1 points

QUESTION 46

1. **Note: Due to a technical problem, the dollar sign in this question is represented as $**

   Given that the following SQL statements have been executed on a PostgreSQL database:

```
CREATE TYPE InventoryItem AS  ( name   text,    supplierID    char(5),
unitPrice    numeric);
CREATE TABLE StockOnHand  ( item  inventoryItem,  quantity  integer);

CREATE FUNCTION stockValue( inventoryItem, integer ) RETURNS numeric AS
 $$

        SELECT $1.unitPrice * $2   AS stockValue;
$$ LANGUAGE SQL;
```

   Which one of the following SELECT statements will produce a list of stock items on hand and the stock value, from the stockOnHand table?

a.

```
SELECT    item.name AS item_name,  item.supplierID AS supplier,
          item.unitPrice AS unit_price, quantity, stockValue(item, quantity)
FROM      stockOnHand;
```

b.

```
SELECT   (stockOnHand.item).name AS item_name,
         (stockOnHand.item).supplierID AS supplier,
         (stockOnHand.item).unitPrice AS unit_price,
         quantity, stockValue(item, quantity)
FROM     stockOnHand;
```

c.

```
SELECT    name  AS item_name, supplierID AS supplier,
          unitPrice   AS unit_price, quantity,
          stockValue(item, quantity)
FROM      stockOnHand;
```

3 points

QUESTION 47

1. State whether the following statement about OODBMSes is TRUE or FALSE:

   The class methods may be stored in the database and then the libraries are dynamically bound to the application at run-time.

   True

   False